

3.1 *Theorem:*

$$\langle p | \hat{T} = \exp\left(\frac{-ibp}{\hbar}\right) \langle p |.$$

Proof. Given

$$\langle x | \hat{T} = \langle x - b |, \quad (1)$$

we can expand with

$$\begin{aligned} \langle p | \hat{T} &= \sum_x \langle p | x \rangle \langle x | \hat{T} \\ &= \int_{-\infty}^{\infty} \exp\left(\frac{-ipx}{\hbar}\right) \langle x - b | dx \end{aligned}$$

and if we let $\xi = x - b$ and subsequently $x = \xi + b$ and $dx = d\xi$,

$$\begin{aligned} \langle p | \hat{T} &= \int_{-\infty}^{\infty} \exp\left(\frac{-ip(\xi + b)}{\hbar}\right) \langle \xi | d\xi \\ &= \int_{-\infty}^{\infty} \exp\left(\frac{-ip\xi}{\hbar}\right) \exp\left(\frac{-ipb}{\hbar}\right) \langle \xi | d\xi \\ &= \exp\left(\frac{-ipb}{\hbar}\right) \int_{-\infty}^{\infty} \exp\left(\frac{-ip\xi}{\hbar}\right) \langle \xi | d\xi \end{aligned}$$

and the second term is an equivalent definition of $\langle p |$ as ξ lies in position-space, simplifying to

$$\langle p | \hat{T} = \exp\left(\frac{-ipb}{\hbar}\right) \langle p |.$$

□

3.2 *Theorem:*

$$\langle p | \hat{D} = \frac{ip}{\hbar} \langle p |.$$

Proof. Given the definition

$$\langle x | \hat{D} = \frac{d}{dx} \langle x |, \quad (2)$$

we can once again expand

$$\begin{aligned} \langle p | \hat{D} &= \sum_x \langle p | x \rangle \langle x | \hat{D} \\ &= \int_{-\infty}^{\infty} \exp\left(\frac{-ipx}{\hbar}\right) \frac{d}{dx} \langle x | dx \end{aligned}$$

Using integration by parts, we can set

$$\begin{aligned} u &= \exp\left(-\frac{ipx}{\hbar}\right) & du &= -\frac{ip}{\hbar} \exp\left(-\frac{ipx}{\hbar}\right) dx \\ dv &= \frac{d}{dx} \langle x | dx & v &= \langle x | \end{aligned}$$

yielding the equation

$$\begin{aligned} \int_{-\infty}^{\infty} \exp\left(-\frac{ipx}{\hbar}\right) \frac{d}{dx} \langle x | dx &= \exp\left(-\frac{ipx}{\hbar}\right) \langle x | \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} -\frac{ip}{\hbar} \exp\left(-\frac{ipx}{\hbar}\right) \langle x | dx \\ &= \exp\left(-\frac{ipx}{\hbar}\right) \langle x | \Big|_{-\infty}^{\infty} + \frac{ip}{\hbar} \int_{-\infty}^{\infty} \exp\left(-\frac{ipx}{\hbar}\right) \langle x | dx \end{aligned}$$

As $\lim_{x \rightarrow \pm\infty} \exp(-ipx/\hbar) = 0$, both terms simplify to

$$\langle p | \hat{D} = \frac{ip}{\hbar} \langle p |.$$

□

3.3 The mirror reflection operator $\hat{R}$ produces the reflection about the origin of any function in x -space. It is called the parity operator in physics. Its x -space instruction is $\langle x | \hat{R} = \langle -x |$. What is its p -space instruction?

Suppose we have a state ψ ,

$$\begin{aligned} \langle p | \hat{R} | \psi \rangle &= \sum_x \langle p | x \rangle \langle x | \hat{R} | \psi \rangle \\ &= \int_{-\infty}^{\infty} \exp\left(-\frac{ipx}{\hbar}\right) \langle -x | \psi \rangle dx \end{aligned}$$

Now let $\xi = -x$, producing

$$\begin{aligned} \langle p | \hat{R} | \psi \rangle &= - \int_{-\infty}^{\infty} \exp\left(\frac{ip\xi}{\hbar}\right) \langle \xi | \psi \rangle d\xi \\ &= - \int_{\infty}^{-\infty} \exp\left(-\frac{i(-p)\xi}{\hbar}\right) \langle \xi | \psi \rangle d\xi \\ &= \langle -p | \psi \rangle \end{aligned}$$

3.4-1 Given $\hat{\tau} |p\rangle = |p + Q\rangle$ where $Q = \hbar/L$. *Theorem:*

$$\langle p | \hat{\tau} = \langle p - Q |.$$

Proof. $\langle p | \hat{\tau} = (\hat{\tau} |p\rangle)^* = \langle p | \hat{\tau}^\dagger = \langle p - Q |$ □

3.4-2 *Theorem:*

$$\langle x | \hat{\tau} = \exp\left(\frac{iQx}{\hbar}\right) \langle x |$$

Proof.

$$\begin{aligned}\langle x | \hat{\tau} &= \sum_p \langle x | p \rangle \langle p | \hat{\tau} = \sum_p (\langle p | x \rangle)^* \langle p | \hat{\tau} \\ &= \sum_p \exp\left(\frac{ipx}{\hbar}\right) \langle p - Q | \end{aligned}$$

Now let $p' = p - Q$,

$$\begin{aligned}\langle x | \hat{\tau} &= \sum_p \exp\left(\frac{i(p' + Q)x}{\hbar}\right) \langle p' | \\ &= \sum_p \exp\left(\frac{ip'x}{\hbar}\right) \exp\left(\frac{iQx}{\hbar}\right) \langle p' | \\ &= \exp\left(\frac{iQx}{\hbar}\right) \sum_p \exp\left(\frac{ip'x}{\hbar}\right) \langle p' | \\ &= \exp\left(\frac{iQx}{\hbar}\right) \langle x | \end{aligned}$$

□

3.5-1 *Theorem:*

$$\langle x | \hat{\tau} \hat{D} | \psi \rangle = \exp\left(\frac{iQx}{\hbar}\right) \frac{d\psi}{dx}$$

Proof. $\langle x | \hat{\tau} \hat{D} | \psi \rangle = \frac{d}{dx} \langle x | \hat{\tau} | \psi \rangle = \exp\left(\frac{iQx}{\hbar}\right) \frac{d}{dx} \langle x | \psi \rangle.$

□

7.1 *Theorem:* $\hat{T} = e^{-b\hat{D}}$

Proof.

$$\langle p | \exp(-b\hat{D}) = \langle p | \exp(-bD(p))$$

where $D(p) = ip/\hbar$ is the eigenvalue of $\hat{D}$ for $\langle p |$, and knowing that

$$\langle p | \hat{T} = \exp\left(\frac{-ibp}{\hbar}\right) \langle p | \quad (3)$$

from problem 3.1,

$$\langle p | \hat{T} = \exp(-bD(p)) \langle p | = \exp(-b\hat{D}) \langle p |,$$

thus

$$\hat{T} = e^{-b\hat{D}}.$$

□

7.2 *Theorem:* $\langle x | \hat{p} | \psi \rangle = \langle x | \hat{p} \hat{T} | \psi \rangle$

Proof. Given that $\hat{T} = e^{-b\hat{D}}$ per problem 7.1, $\langle x | \psi \rangle$ and $\langle x | \hat{T} | \psi \rangle$ differ by only the phase $e^{-b\hat{D}}$ which cannot have any effect on the probability as $\forall \theta, |e^{i\theta}|^2 = 1$. $\square$

7.3 *Theorem:* $\forall \psi, \forall t, |\langle E | \psi \rangle|^2 = |\langle E | \hat{U}_t | \psi \rangle|^2$.

Proof. $\hat{U}_t = \exp\left(\frac{-it\hat{H}}{\hbar}\right)$ is a unit vector on the complex plane, so it cannot have any effect on the probability. $\square$

7.4 What is the characteristic frequency $\Omega/2\pi$ for a neutron in a magnetic field of 1 kG? In what region of the electromagnetic spectrum is this: radio, microwave, infrared, visible, or x-ray?

The characteristic frequency is

$$\omega = \frac{B}{2\pi} |\Omega| = \frac{B}{2\pi} \left| \frac{g_n e}{2m_n} \right| = \frac{1 \times 10^3 \text{ kG}}{2\pi} \left| \frac{(-3.83)(1.60 \times 10^{-19} \text{ C})}{2(1.67 \times 10^{-27} \text{ kg})} \right| \simeq 2.92 \text{ MHz}$$

and this is a radiowave.

7.5 *Theorem:* $\hat{U}_t |\mu = 1/2\rangle = \frac{\hat{U}_t}{\sqrt{2}} (|m = 1/2\rangle + i |m = -1/2\rangle)$ where μ and m are eigenvalues of $\hat{S}_y$ and $\hat{S}_z$ respectively.

Proof. The unit time transform can be removed from both sides, so the proof is just

$$|\mu = 1/2\rangle = |m = 1/2\rangle + i |m = -1/2\rangle$$

and this can be proven using the change-of-basis matrix from $\text{eig}_\lambda \hat{S}_y$ to $\text{eig}_\lambda \hat{S}_z$ $\square$